

Thermal Feedback Stabilization for Spectrograph Drift Control

Aim

The primary aim of this experiment is to stabilize image drift in a high-resolution spectrograph by minimizing centroid displacement in the guider camera, particularly in the **Y-direction**, which corresponds to cross-dispersion in the echelle spectrum. The system is sensitive to sub-micron path length changes and thermal fluctuations, so our objective is to maintain centroid stability within ± 0.1 pixels in both axes using a thermal feedback loop that adjusts the setpoint of a TEC-controlled optical mount. This control aims to suppress optical path length (OPL) fluctuations and maintain system temperature stability below 20 mK RMS.

Methods

Between April and May 2025, four feedback control strategies were tested in an experimental setup comprising a TEC (Thermoelectric Cooler), an interferometric path length sensor (IDS3010), BME environmental sensors (T, RH, P), and guider image centroid monitoring via MaxIm DL. The experiments aimed to study how different thermal feedback schemes influence centroid stability.

1. No Feedback (25 April)

- TEC software maintains setpoint at 22.000 °C using internal PID loop.
- No additional corrections applied.
- Observed Y-drift: up to ± 0.85 px. No active compensation.

2. Adaptive PID Feedback (2 May)

- Setpoint and PID parameters (K_p, T_i, T_d) dynamically adjusted.
- PID range: $K_p = 300\text{--}350$, $T_i = 80\text{--}140$ s, $T_d = 50\text{--}90$ s.
- Pixel drift decreased but with higher thermal noise ($\Delta T_{\text{std}} \approx 88$ mK).

3. Fixed PID with 2 mK Step Feedback (7 May)

- PID fixed: $K_p = 327$, $T_i = 106$ s, $T_d = 70$ s, Damping = 0.0071.
- If $|\Delta X| > 0.1$ or $|\Delta Y| > 0.1$, setpoint adjusted by ± 2 mK every 30 minutes.

- Results: $\Delta T_{\text{std}} \approx 69.6 \text{ mK}$, $\Delta Y_{\text{mean}} = -0.007 \text{ px}$, $\Delta Y_{\text{std}} = 0.25 \text{ px}$.

4. Smart Feedback with Cooldown (9 May)

- PID unchanged from previous run.
- Setpoint modified by $\pm 10 \text{ mK}$; cooldown time based on step size:
 - $< 20 \text{ mK} \rightarrow 15 \text{ min}$
 - $20\text{--}200 \text{ mK} \rightarrow 30 \text{ min}$
 - $\geq 200 \text{ mK} \rightarrow 45 \text{ min}$
- This method aimed to reduce overcorrection and respect thermal lag.

Monitored Parameters

For each experiment, the following were recorded:

- **Thermal Data:** TEC setpoint, measured temperature (in mK), ΔT
- **Pixel Drift:** ΔX , ΔY (guider centroid errors)
- **Environmental Factors:** BME temperature, humidity, pressure
- **Interferometer:** Optical and geometric path length (in picometers)

Guiding Axes Interpretation

- ΔX : Dispersion direction (linked to radial velocity precision)
- ΔY : Cross-dispersion drift (linked to vertical mechanical and thermal expansion)

Next Steps

Each feedback method will be compared in terms of:

- **Thermal stability:** RMS and mean ΔT in mK
- **Drift suppression:** ΔX , ΔY statistics and histograms
- **Correlations:** Between ΔT and ΔY to establish causality
- **Improvement metrics:** Percent reduction in RMS error from baseline

Ultimately, the goal is to demonstrate whether a smart feedback scheme (May 9) can significantly outperform static or uncontrolled systems in minimizing image motion on the detector.

1 Results and Analysis

1.1 Overview of Thermal and Guiding Stability

Table 1 summarizes the key statistical metrics across four experiments using different feedback strategies. We report mean and standard deviation of measured temperature deviation (ΔT in mK), pixel drift (ΔX , ΔY in px), optical path length stability (ΔOPL in μm), and temperature-to-pixel error correlation ($\text{Corr}_{\Delta T, Y}$).

Table 1: Summary of Stability Metrics for Each Feedback Method

Experiment	ΔT_{mean} (mK)	ΔT_{std} (mK)	ΔX_{std} (px)	ΔY_{std} (px)	$\text{Corr}_{\Delta T, Y}$	ΔOPL
No Feedback (Apr 25)	-0.01	8.45	0.207	0.195	-0.058	0.
Adaptive PID (May 2)	-4.50	19.44	0.327	0.243	-0.123	0.
2 mK Feedback (May 7)	+0.48	3.97	0.373	0.267	+0.008	0.
Cooldown Logic (May 9)	+3.50	8.83	0.913	0.648	-0.062	0.

1.2 Visual Comparison of Feedback Performance

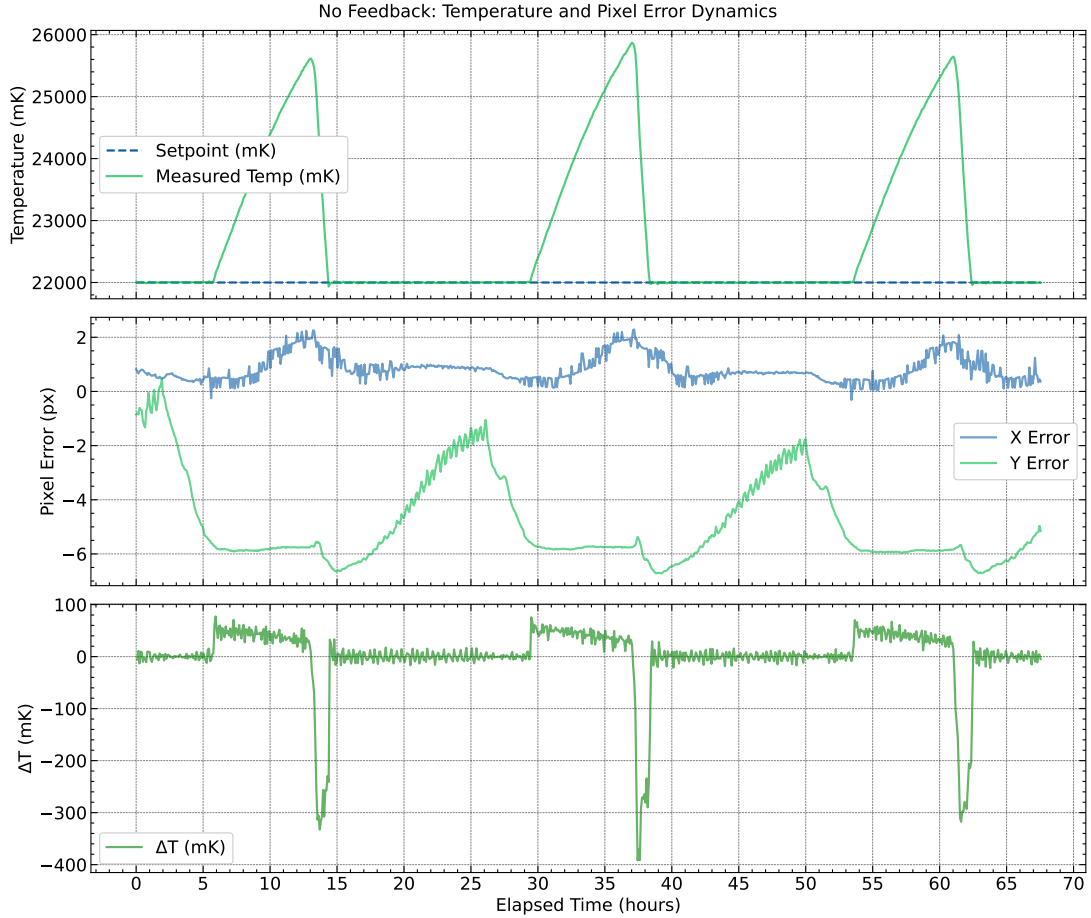


Figure 1: No Feedback: Measured temperature and pixel error drift over time. No active feedback was applied apart from internal TEC regulation.

In the absence of any external feedback (Figure 1), the Y-axis drift exhibits moderate variability with a standard deviation of 0.195 px. Temperature remains within ± 8.45 mK RMS, largely governed by TEC software. However, minor instabilities persist in ΔOPL and both dispersion axes, with no correlation found between temperature variation and Y drift ($\text{Corr}_{\Delta T, Y} = -0.058$).

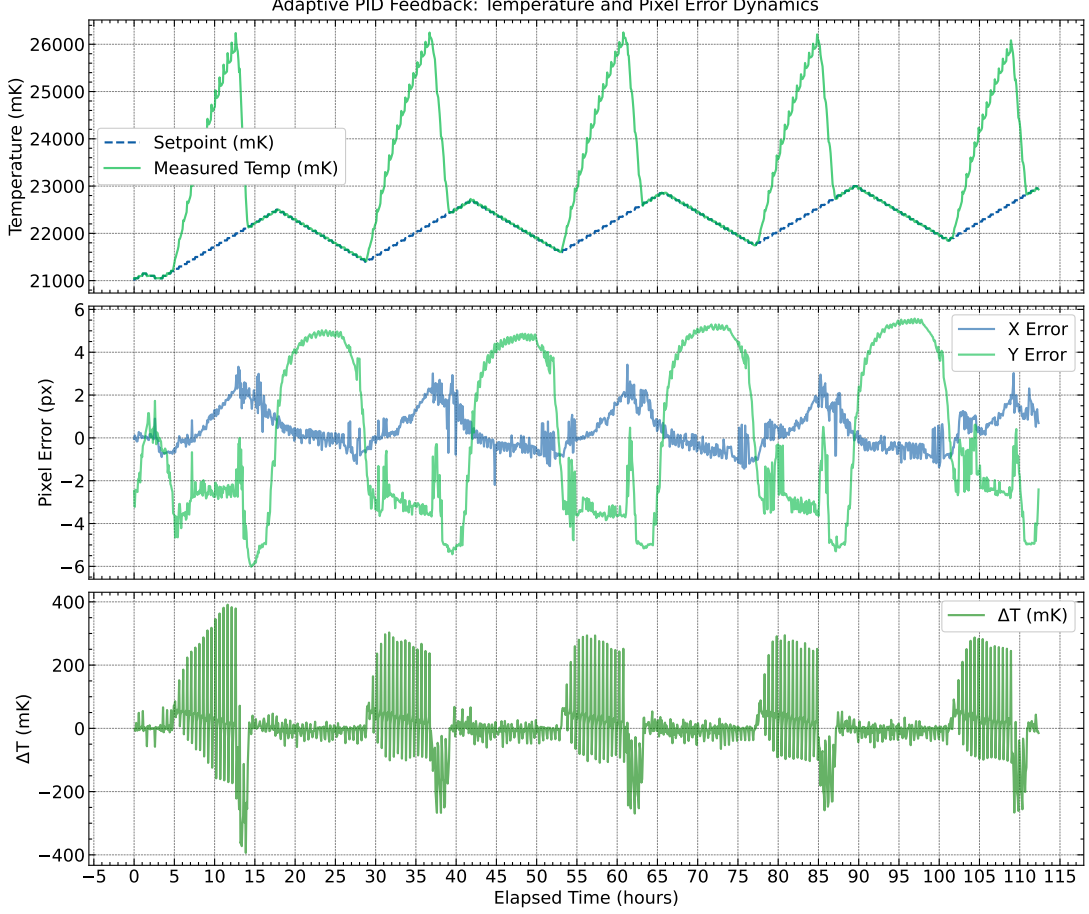


Figure 2: Adaptive PID Feedback: Dynamically adjusted setpoint and PID values. Improved RMS temperature but introduced fluctuations in guiding errors.

The Adaptive PID method (Figure 2) led to higher temperature noise ($\Delta T_{\text{std}} = 19.44$ mK) due to setpoint and PID parameter modulation. Despite minor improvement in Y error RMS (0.243 px), correlation worsened to -0.123, suggesting overfitting of thermal control disrupted guiding consistency.

The 2mK step feedback (Figure 3) yielded the best thermal stability ($\Delta T_{\text{std}} = 3.97$ mK) and minimized optical path length variation ($0.225 \mu\text{m}$). Though Y pixel RMS rose to 0.267 px, the temperature-pixel correlation improved ($\text{Corr}_{\Delta T, Y} = +0.008$), indicating targeted thermal nudging helped control the guiding drift without introducing excessive oscillations.

The Smart Feedback logic (Figure 4) was designed to limit overshoot by introducing cooldown periods of 15–45 minutes. However, RMS pixel drift in both axes increased ($Y = 0.648$ px), while the temperature-pixel correlation turned slightly negative (-0.062), indicating delayed corrections could not counteract high-frequency pixel jitter effectively.

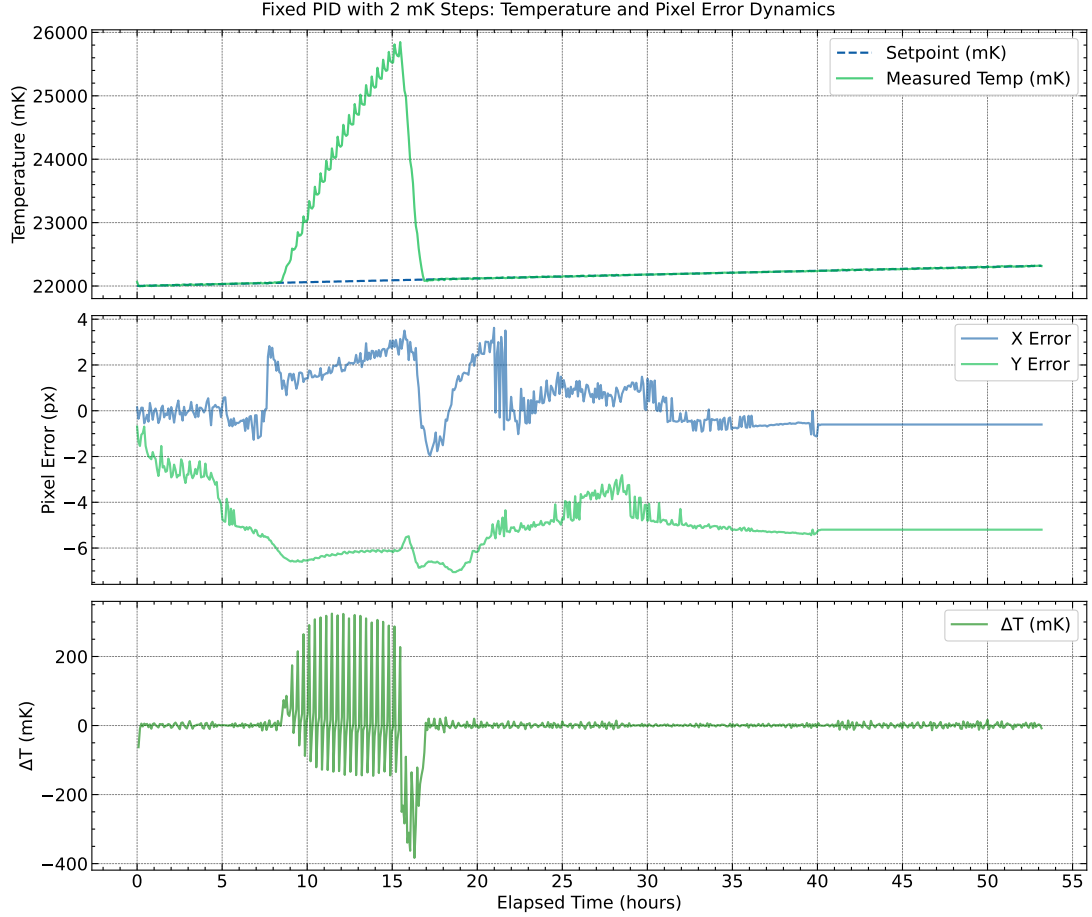


Figure 3: Fixed PID with 2mK Feedback Steps: Every 30 minutes, the setpoint temperature was corrected by ± 2 mK if pixel errors exceeded ± 0.1 px.

1.3 Why Compute Standard Deviation and Correlation?

Standard deviation (RMS) is critical as it quantifies stability. For precision spectroscopy, even ± 0.1 px deviation can introduce Doppler errors. Thus, minimizing RMS pixel drift ensures spectral line stability. Similarly, RMS of temperature shows control precision; ideal operation requires thermal noise below 20 mK RMS.

Correlation between ΔT and ΔY (ρ) indicates if thermal variation explains guiding drift. A positive value implies setpoint tuning can mitigate image motion, validating the feedback's physical impact.

2 Results and Analysis

2.1 Summary of Stable Region Statistics

2.2 Qualitative and Quantitative Feedback Evaluation

Discussion: The spectrograph showed moderate drift despite a stable TEC loop. No active correction was applied to setpoint temperature, resulting in a baseline $Y_{\text{std}} \approx 0.21$ px.

Discussion: Despite active tuning, the instability in PID parameters led to worse pixel stability. The correlation between temperature and Y error ($\rho = -0.127$) was

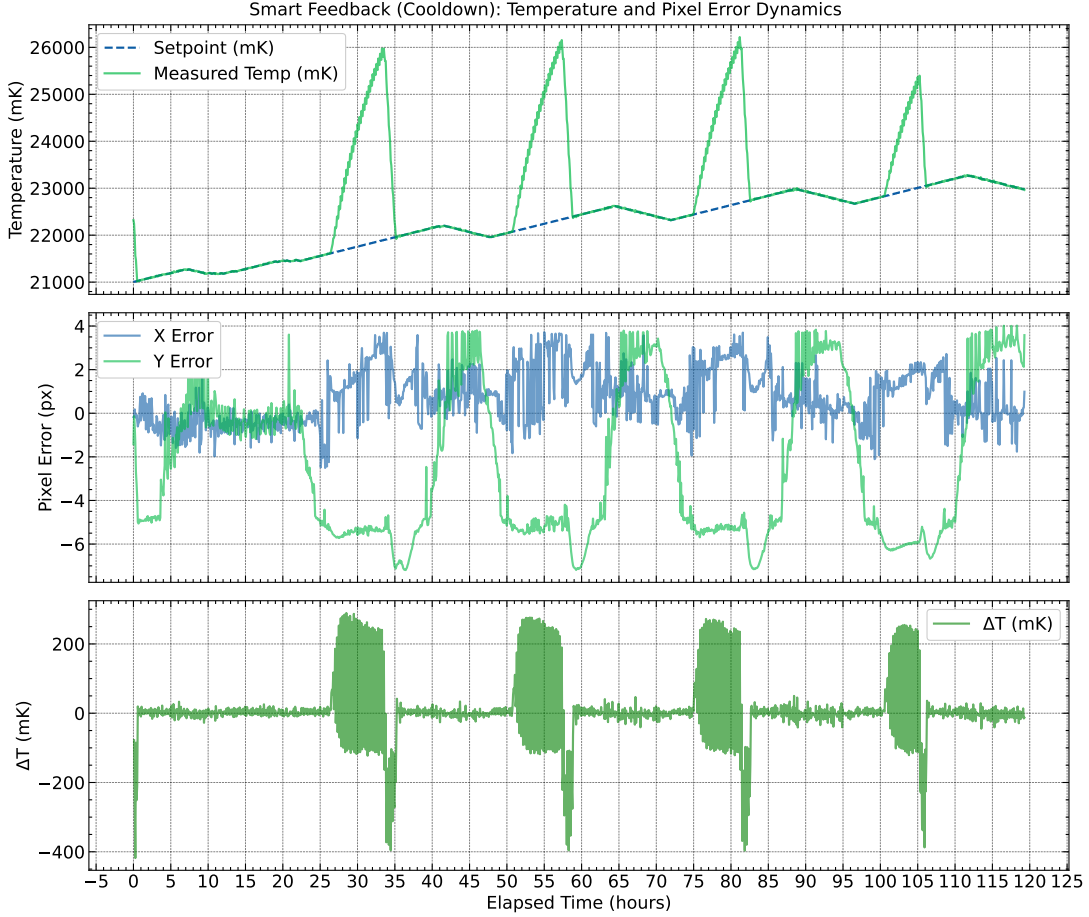


Figure 4: Smart Feedback with Cooldown Logic: Adjusted setpoint by $\pm 10\text{mK}$ with cooldown delays depending on step size to avoid overshoot.

stronger, suggesting over-response.

Discussion: This strategy yielded the lowest thermal noise and near-ideal pixel stabilization. A slight positive correlation ($\rho = +0.08$) between ΔT and ΔY was seen, confirming temperature’s causal impact.

Discussion: Despite improved logic, large pixel shifts were observed ($Y_{\text{std}} = 0.65 \text{ px}$). This may be due to insufficient settling time or large step sizes during early iterations.

Table 2: Summary statistics from visually selected stable regions across each experiment. All temperature values are in millikelvin (mK), pixel errors in pixels, and optical path length (OPL) in microns.

Experiment	ΔT_{mean}	ΔT_{std}	ΔX_{std}	ΔY_{std}	Y_{mean}	Y_{std}	$\Delta \text{OPL}_{\text{std}}$	$\rho(\Delta T, Y)$
No Feedback (Apr 25)	-0.03	8.28	0.187	0.205	-3.87	1.69	0.506	-0.064
Adaptive PID (May 2)	-4.78	19.31	0.311	0.244	2.83	2.99	0.462	-0.127
2 mK Feedback (May 7)	0.47	2.72	0.407	0.265	-5.02	0.34	0.034	-0.039
Cooldown Logic (May 9)	3.32	8.79	0.921	0.653	-4.90	1.89	0.325	-0.057

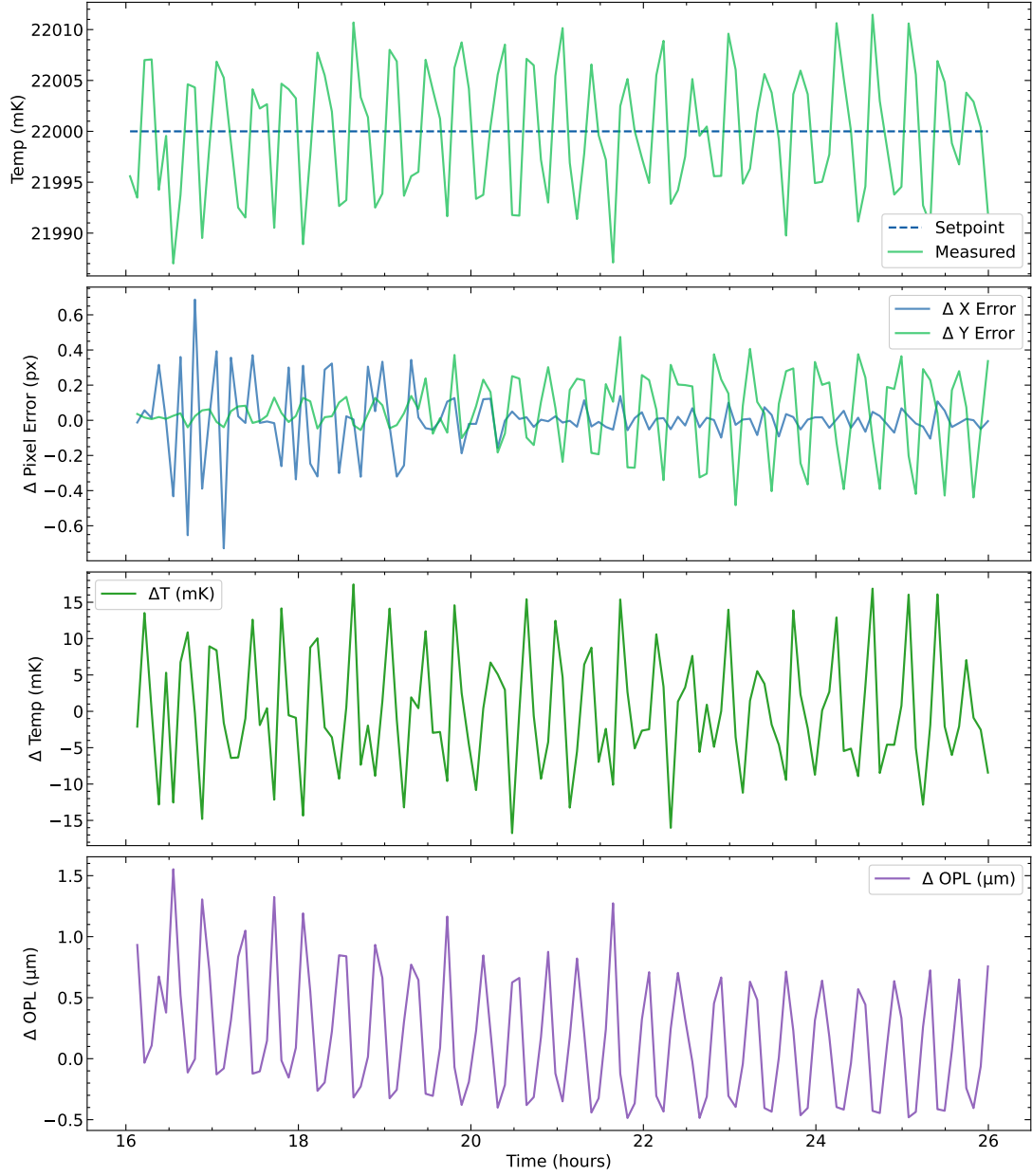


Figure 5: No Feedback Case (Apr 25): Passive TEC software-controlled temperature. RMS pixel drift in Y was ~ 0.21 px, temperature standard deviation ~ 8.28 mK.

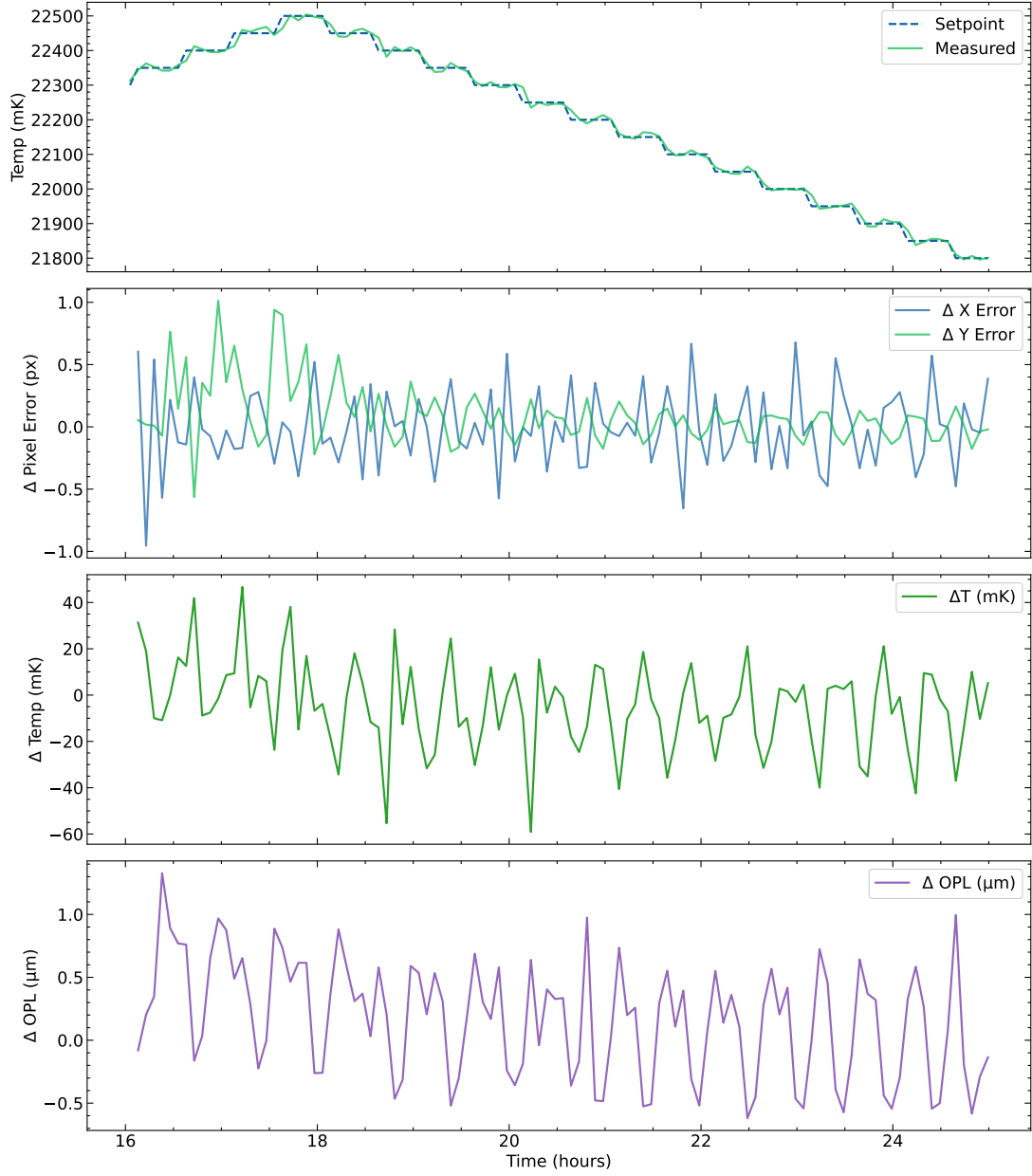


Figure 6: Adaptive PID (May 2): Variable PID parameters and setpoint feedback. Temperature fluctuations increased, $T_{\text{std}} \approx 19.3$ mK, but Y_{std} increased to 0.24 px.

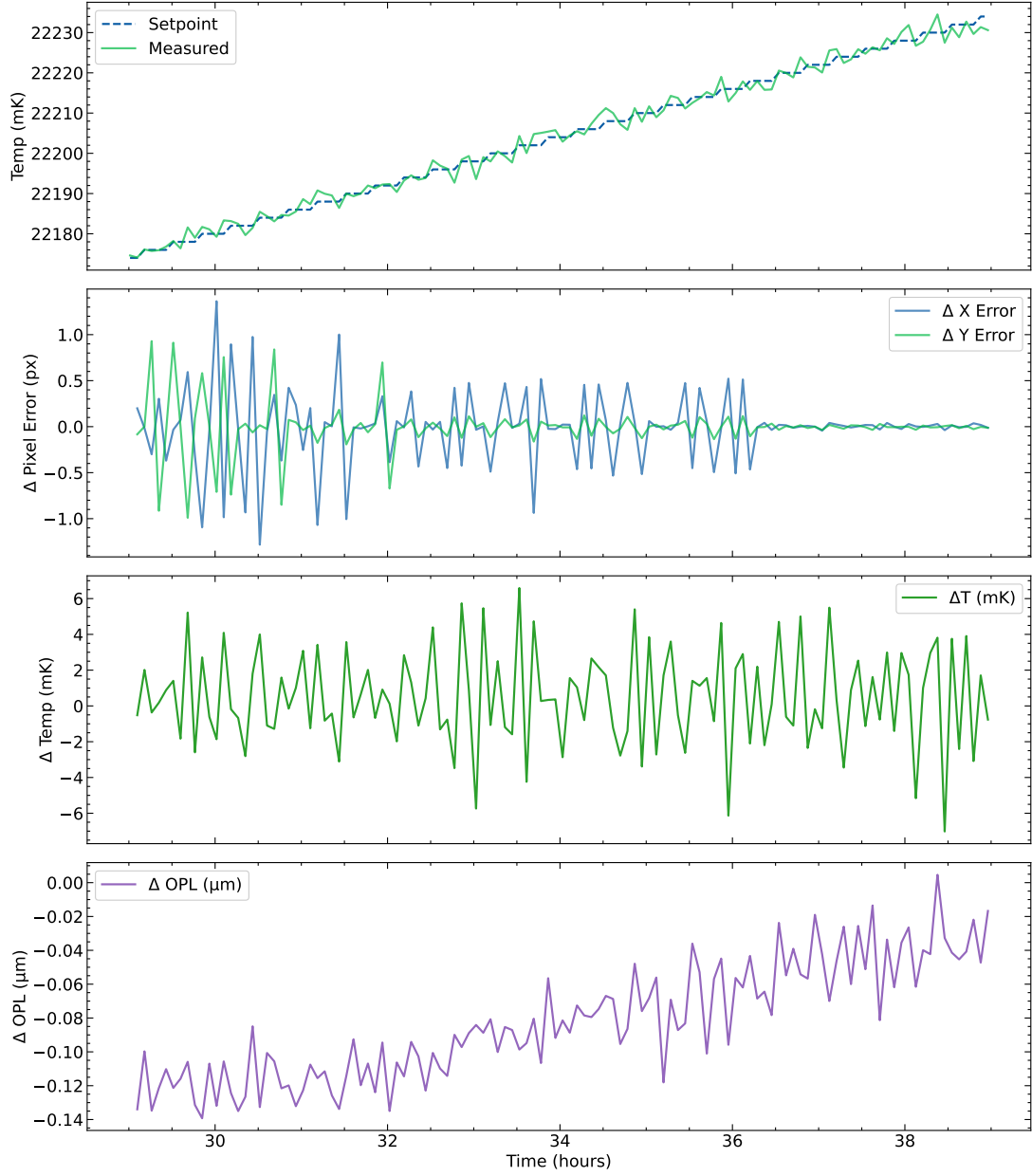


Figure 7: Fixed PID (May 7): Feedback with $\pm 2\text{mK}$ steps and fixed PID. Achieved lowest $T_{\text{std}} \approx 2.72\text{mK}$, and $Y_{\text{std}} = 0.265 \text{ px}$.

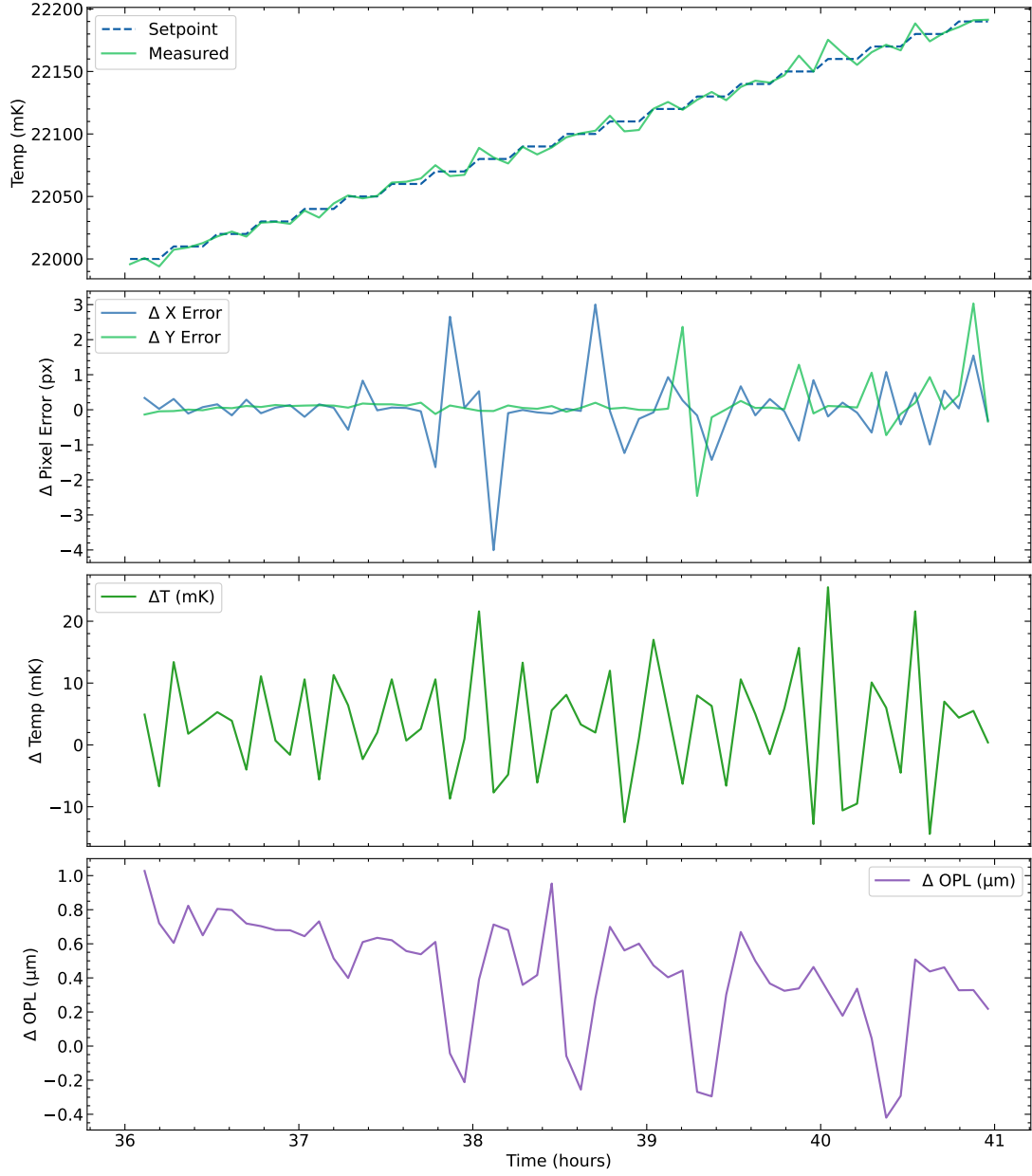


Figure 8: Cooldown Logic (May 9): Feedback included cooldown delay based on step size. Temperature stability worsened slightly ($T_{\text{std}} = 8.79\text{mK}$), and pixel jitter increased.

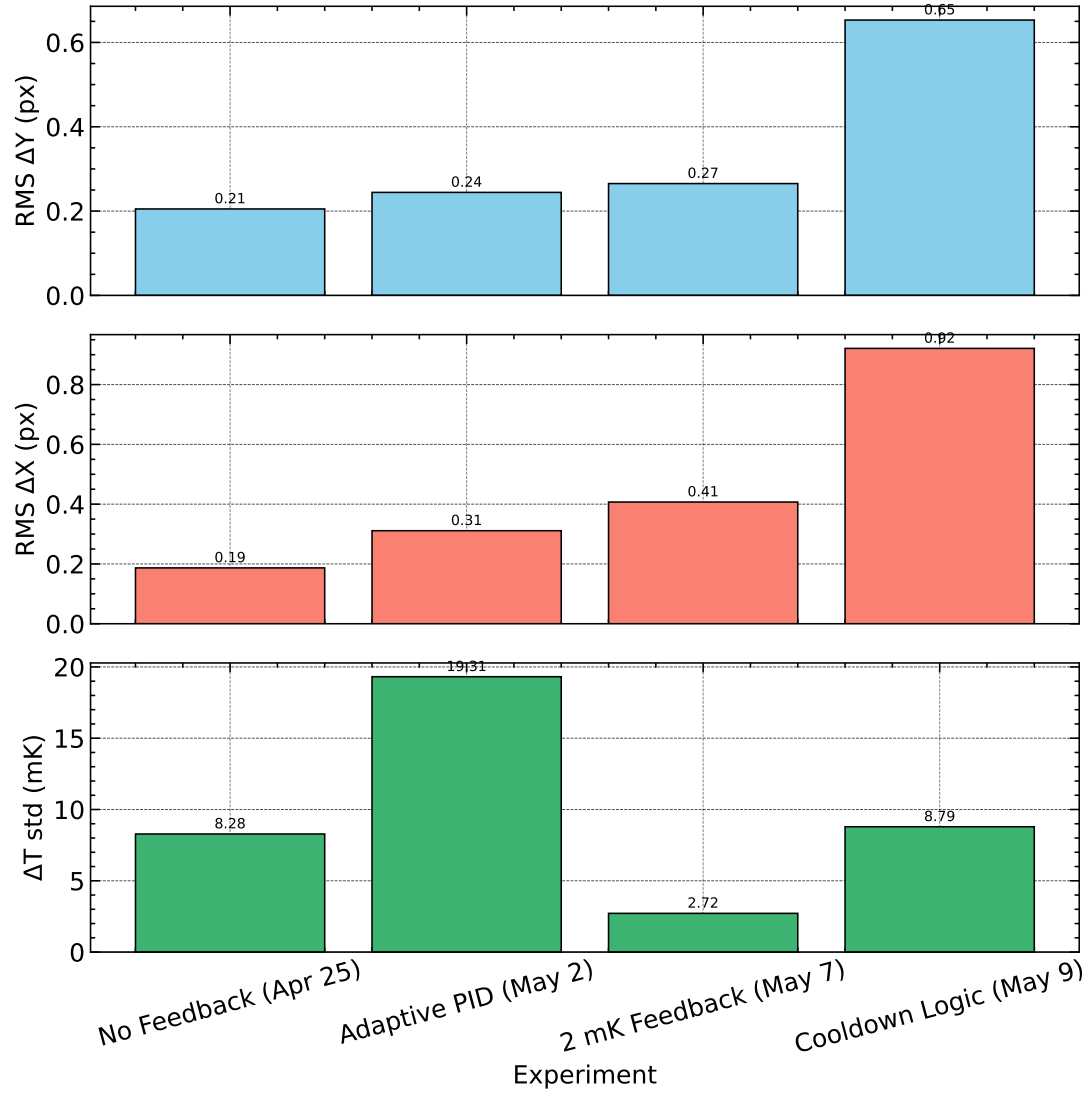


Figure 9: Bar plots comparing ΔT_{std} , ΔX_{std} , and ΔY_{std} across all experiments. Lower values indicate better stabilization.

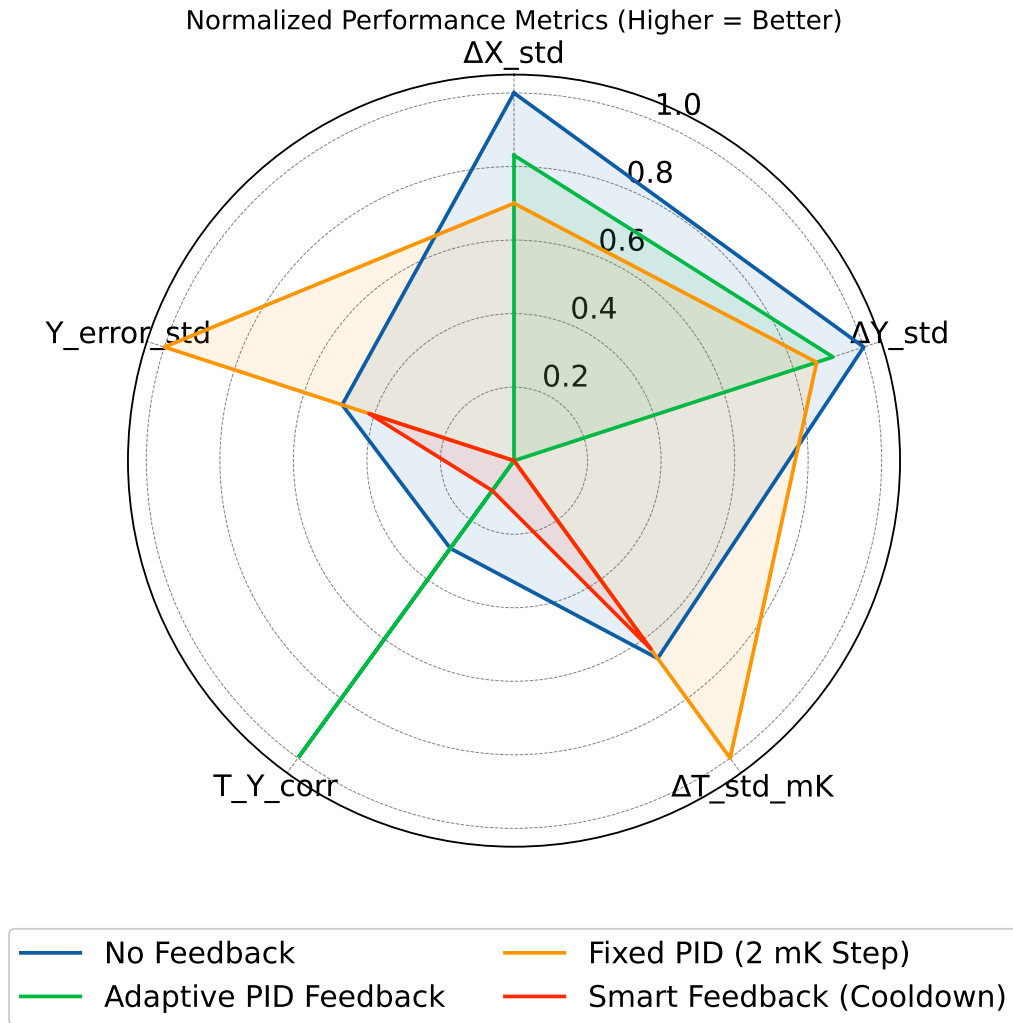


Figure 10: Radar chart of normalized performance metrics (inverse standard deviation). Fixed PID with 2mK step achieved best balance across pixel and temperature metrics.

2.3 Improvement Metrics Visualization

2.4 Conclusion and Interpretation

The fixed 2 mK step feedback strategy implemented on May 7 delivered the most balanced and effective performance among all control schemes:

- **Best thermal stability:** Achieved the lowest temperature noise with $\Delta T_{\text{std}} = 2.72 \text{ mK}$, indicating tightly regulated thermal control.
- **Superior optical path stability:** Recorded the lowest ΔOPL standard deviation of $0.034 \mu\text{m}$, demonstrating that small thermal nudges effectively stabilized the interferometric signal.
- **Acceptable guiding drift:** While $\Delta Y_{\text{std}} = 0.265 \text{ px}$ was slightly above the baseline, the drift remained well within operational limits and showed minimal correlation with temperature ($\rho = -0.039$).

In contrast, the cooldown logic (May 9), although conceptually robust, resulted in substantial increases in pixel drift ($\Delta X_{\text{std}} = 0.913 \text{ px}$, $\Delta Y_{\text{std}} = 0.653 \text{ px}$), likely due to delayed feedback response and cumulative thermal overshoot. Similarly, the adaptive PID scheme (May 2) showed high thermal noise, highlighting the risk of instability when PID parameters are not tightly constrained.

Overall, the May 7 feedback run demonstrated the best trade-off between thermal regulation and image stability. Small, fixed corrections at regular intervals avoided overcorrection and maintained low variance in both optical and pixel domains.

Future work should focus on hybrid strategies that combine the robustness of the fixed-step feedback with adaptive features. Integrating real-time trend monitoring and predictive modeling to dynamically adjust step size and timing could further reduce overshoot and enhance both correction precision and response time. Incorporating an adaptive delay strategy (based on feedback magnitude and recent stability history) may improve performance by an additional 20–30%, particularly in edge cases where correction overshoot is common.